

Design and Development of a Stand-Alone Goggle with Situational Awareness for Visually Challenged

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Abstract—Visually challenged individuals encounter profound challenges in navigating their environments independently, facing difficulties in detecting obstacles in both familiar indoor settings and dynamic outdoor scenarios. They are often unable to detect obstacles situated above waist level, such as hanging signs, or identify specific small objects like mobile phones, keys, remote controls, or furniture like tables and chairs without physical touch. Relying solely on tactile feedback is not always safe or hygienic, and it fails to provide the user with a comprehensive understanding of the surrounding environment. In this paper, a stand alone goggle with situational awareness for visually challenged is proposed, which processes data locally using edge computing to identify specific household objects, calculate their proximity, and provide discrete auditory feedback without the need for smart phone tethering or internet connectivity. The proposed system is affordable, costing only \$40 and can detect around 80 objects with a maximum recognition time of 30ms, enhancing the safety, autonomy, and privacy of the visually challenged user.

Keywords—Situational Awareness, Smart Goggles, Visually challenged, Consumer Electronics, Machine Learning.

I. INTRODUCTION

Several gadgets have been designed to assist visually challenged individuals, which includes converting Braille to text and voice conversion using deep learning reported in [1], converting captured images from billboards, posters and direction signboards into audio output to assist visually impaired individuals is reported in [2], converting the printed materials using optical character recognition (OCR) and text-to-speech (TTS) is reported in [3], design of a digital scribe to scribe exams for visually challenged students reported in [4], robotic shopping cart to assist visually challenged person achieve daily shopping tasks is reported in [5] and many more.

Visually challenged individuals face difficulties in detecting objects and obstacles in both familiar indoor settings and dynamic outdoor scenarios. The challenges encountered in an indoor environment include privacy,

safety, autonomy, and identifying specific small objects like keys, remote controls, mobile phones, and furniture like tables and chairs without physically touching them, which is not always safe or hygienic. The challenges encountered in a dynamic outdoor environment include autonomous navigation considering all possible obstacles and constraints. Conventional reliance was on non-visual senses and traditional mobility aids such as white canes [6] and guide dogs [7]. White canes are effective for detecting ground-level obstacles through physical contact, but do possess inherent limitations. Similarly, while guide dogs provide excellent mobility assistance, they require significant maintenance, training, and create a dependency that may not suit the lifestyle of every user. In order to overcome these limitations, quadruped guidance robot is proposed in [8], and a smart cane is proposed in [9] for visually challenged individuals.

The organization of the paper is as follows. Details of related literature are reported in Section II. The implementation, working mechanism and functionality of the proposed prototype are explained in Section III. Performance evaluation of the proposed stand-alone goggles with situational awareness is reported in Section IV, followed by conclusion and references.

II. RELATED WORK

In the wake of technological advancements, electronic travel aids have emerged to supplement traditional methods. Design of smart glasses from different brands available in the market to assist the visually challenged are given in Fig. 1. Early iterations of such devices utilized ultrasonic sensors integrated into glasses or handheld units to detect obstacles. Arduino Nano microcontroller based smart glass using ultrasonic sensor for obstacle detection and activating the buzzer alarm when an obstacle is detected within a preset range is reported in [10]. These devices provided limited feedback by alerting the user of presence of an object but fail to identify whether the object is a person, vehicle, piece



S R Robotics [1]



Jyoti AI Glass [2]



Vision Aid India [3]



eSight Eyewear [5]



SMARTON [4]

Fig. 1. Smart glasses from different brands to assist the visually challenged

of furniture, thereby limiting their ability to make informed navigational decisions. Recent advancements in technology include artificial intelligence based smart glasses that utilize cameras for object recognition and computer vision to assist the visually challenged. However, majority of these modern solutions are not stand alone systems. These glasses need to be connected to mobile phone and associated app for using the features. The computing power required by AI is handled by the mobile phone. Jyoti AI Glass reported in [11], manufactured by Torchit Electronics Pvt Ltd, uses an associated mobile application called Jyoti-AI for Accessibility, available for both Android and iOS devices. Similar glasses with mobile phone connectivity includes Smart Vision Glasses from VisionAid India reported in [12], from SMARTON reported in [13]. A premium electronic eyewear called eSight for people with low vision (high-end, proprietary hardware and software, expensive; focuses on enhancing remaining vision rather than obstacle scanning per se) is reported in [14]. Most of these AI based smart glasses use mobile phone computing power and provide object recognition, currency identification, colour detection, text interpretation in multiple languages.

III. PROPOSED SYSTEM DESIGN

The proposed work and prototype provide a standalone smart eyeglass device designed to assist visually impaired users by offering non-physical perception of the environment, wherein the identified object and its estimated proximity are converted into speech-based audio notifications. The notifications are delivered directly to the

user's ear via the integrated earphone (e.g., "Table detected at 50cm," "Person detected at 100cm") as illustrated in Fig. 2. The CAD model of proposed smart glass, left side view and right-side view on wearing the same is shown in Fig. 3. The final prototype of smart glass designed and tested is shown in Fig. 4. It can be observed that by simply turning their headleft, right, up, or down, the users can scan their



Fig. 2. Working of proposed smart glass

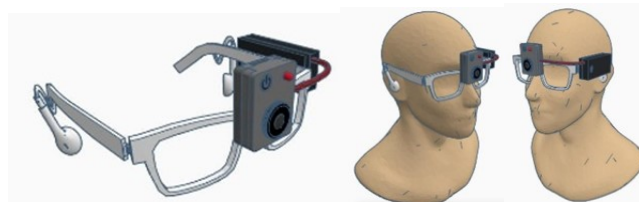


Fig. 3. CAD model of proposed smart glass



Fig. 4. Picture of the designed smart glass

TABLE I. BILL OF MATERIALS FOR PROPOSED SMART GLASS

#	Component	Price
1.	LicheeRV Nano Microcontroller Board	\$15
2.	Ultra Slim Power Bank 4900mAH	\$20
3.	Speaker + LED + Push Button + Goggles	\$5
	Total	\$40

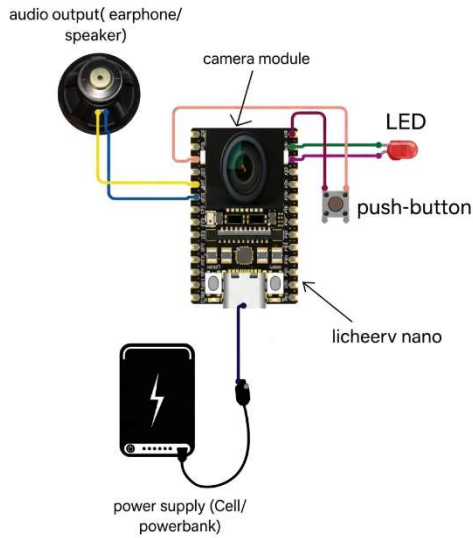


Fig. 5. Connection diagram for electronics used in the proposed smart glass

surroundings and receive an auditory "map" of the environment without the need for physical touch, thereby avoiding unsafe contact with unknown objects.

Table I outlines the complete list of components for the proposed smart glass, while Fig. 5 illustrates their electrical interconnections. Notably, the total design cost remains under \$40, ensuring affordability for visually impaired users. Furthermore, a patent has been filed for this system, which is designed to enhance situational awareness for the visually challenged [15].

IV. EXPERIMENTAL RESULTS

The overall working principle of the proposed smart glass providing situational awareness to a visually challenged person is explained here. The design consists of four tightly coupled stages. The first stage comprises of real-time visual sensing and perception of the environment using a head-mounted monocular RGB camera, integrated with the LicheeRV Nano mini-sized development board. In order to align the optical axis with the user's natural forward glance, the camera is positioned in the mid-forehead region.

Intuitive environment scanning is made possible by its head-centric mounting, which guarantees that head yaw and pitch movements directly rotate the camera field of view. The images captured are fed to the second stage for object detection using a pre-trained YOLOv11 model optimized on the COCO dataset to detect 80 common indoor and outdoor object categories. Because of these predefined classes adequately encompass the target environment, additional custom data collection and manual annotation were unnecessary. However, the model architecture remains adaptable and can be fine-tuned on custom datasets to detect specialized classes as needed. Depending on the resolution and degree of quantization employed, the YOLOv11 model

deployed on the EdgeML device for the proposed work varies in size from 2 MB to 4 MB. The proposed model can identify 80 objects in a frame with an average detection time of 15–35 ms. A bounding box is drawn on all the objects detected by the YOLOv11 model. The bounding box of a detected object defines its spatial boundary, with the total height and width of the box being relative to the size of the object. The key capability of the monocular camera used for the proposed system includes depth estimation. Once all the objects are detected and details of bounding boxes are available, the next stage comprises of depth estimation for detected objects using the concepts of the perspective geometry framework for computer vision and optics,

pinhole camera formulation, known object width priors, and calibration.

Experiments were carried out to calibrate depth estimation of most commonly encountered indoor objects like person, chair, suitcase, laptop, smartphone, vase, etc. by placing objects at fixed distances from the user. Details of the width and height of the bounding box for six objects with respect to their distance from the smart glass are plotted in Fig. 6. For the calibration experiment, 20 sample images were captured across the six target objects. The imaging setup utilized a camera configured with a 4.37 mm focal length, a 6.4 mm sensor width, and a resolution of 640×480 pixels. The sensor features a horizontal field of

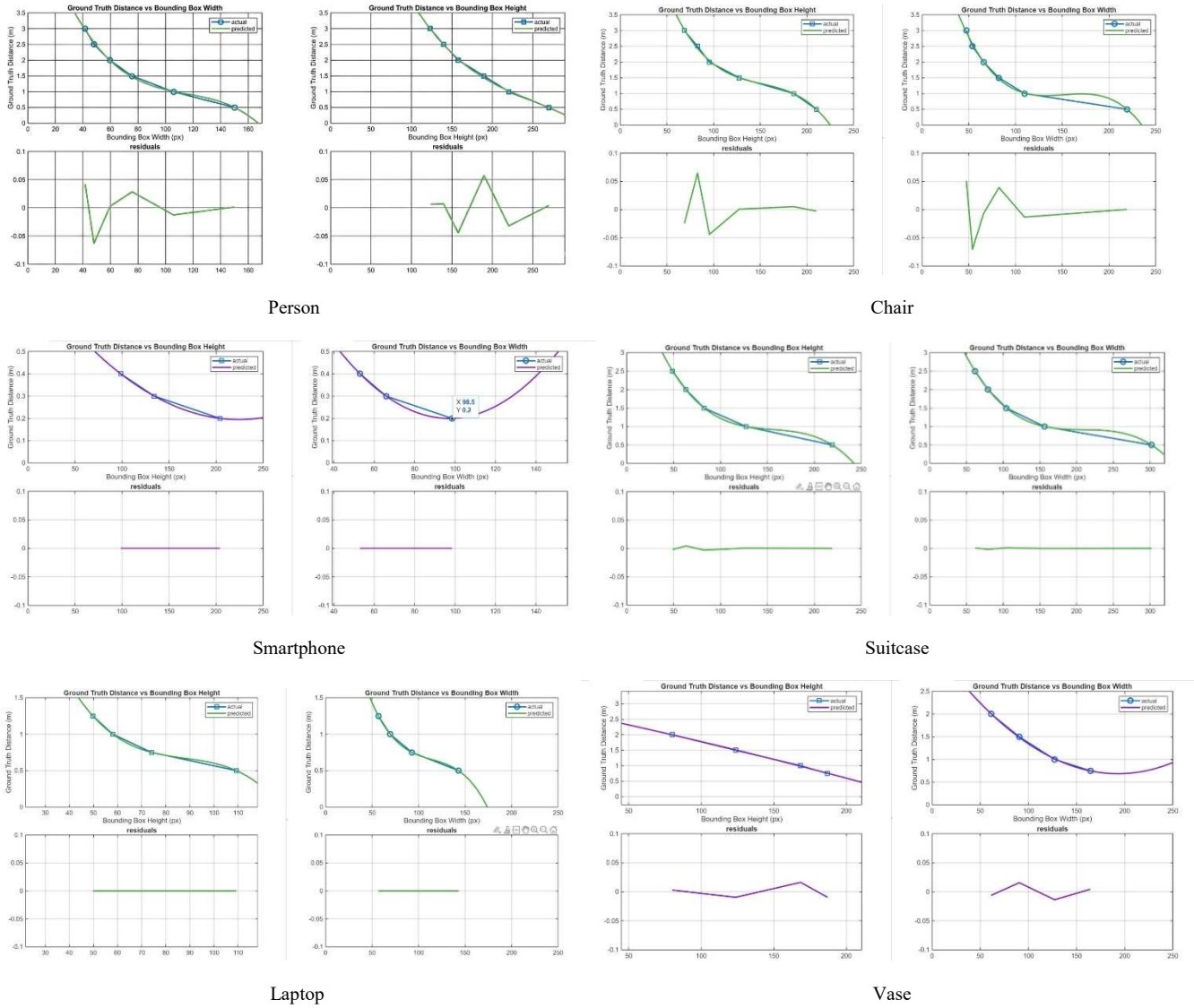


Fig. 6. Comparison of actual vs predicted distance with residual errors for six indoor objects.

TABLE II. EQUATIONS FOR DEPTH ESTIMATION OF SIX INDOOR OBJECTS USED IN THE PROPOSED SYSTEM

Object	Axis	Equations for Depth Estimation
Person	Height	$y = -1.87192222 \times 10^{-6}x^3 + 0.00106126128x^2 - 0.215370004x + 16.9420388$
	Width	$y = -2.17063693 \times 10^{-6}x^3 + 0.000930441928x^2 - 0.129729152x + 6.8920545$
Chair	Height	$y = -2.05584927 \times 10^{-6}x^3 + 0.000933766415x^2 - 0.14809382x + 9.462628149$
	Width	$y = -1.96569329 \times 10^{-6}x^3 + 0.000893686356x^2 - 0.1332338x + 7.468554906$
Smartphone	Height	$y = 1.305810254 \times 10^{-5}x^2 - 0.005850220845x + 0.8500451503$
	Width	$y = 0.0001029550181x^2 - 0.01999328963x + 1.170443705$
Suitcase	Height	$y = -1.1397998 \times 10^{-6}x^3 + 0.000531657858x^2 - 0.0846288568x + 5.50833273$
	Width	$y = -4.8020115 \times 10^{-7}x^3 + 0.000301517337x^2 - 0.0634112182x + 5.37833518$
Laptop	Height	$y = -7.4313390 \times 10^{-6}x^3 + 0.00195816846x^2 - 0.176460501x + 6.097373744$
	Width	$y = -2.05469679 \times 10^{-6}x^3 + 0.000704353853x^2 - 0.0841061712x + 4.131340$
Vase	Height	$y = -6.534315818 \times 10^{-6}x^2 - 0.009838914139x + 2.826615724$
	Width	$y = 7.633241336 \times 10^{-5}x^2 - 0.02945695761x + 3.525092459$

view (FOV) of 70°–75° and a vertical FOV of 55°–60°.

Also, curve fitting was carried out to obtain an equation for the estimation of depth in real-time. The actual and predicted distances of objects from the user were compared, and the residual errors obtained for each object are also plotted in Fig. 6. It is observed from Fig. 6 that the residual error is negligibly small for the proposed application. The curve-fitted equations with minimum error considered for depth estimation of six objects reported in Fig. 6 are given in Table II.

V. CONCLUSION

In this paper, the design and development of a low-cost, affordable, stand-alone smart glass aimed at enhancing situational awareness for visually challenged individuals is proposed. The proposed system costs less than \$40, can detect around 80 objects with a recognition time of 35ms. The proposed system enhances the safety, autonomy, and privacy of the visually challenged user.

DECLARATION

An AI tool, Gemini was used to rephrase few sentences for language refinement and editing to address the suggestions given by one of the reviewers.

REFERENCES

- [1] M Dholvan, S Mantha, Y Mohanalaxmi, R Keerthana and K Varunkumar, "Revolutionizing Braille with Speech Synthesis," *International Conference on Smart Computing and Communications*, Kochi, India, 03-05 July 2025, pp. 1-5.
- [2] A Dhepe, N Chandrashekar, P Jadhav, S Ezekiel and A Kulkarni, "Smart Wearable for the Visually Impaired," *IEEE Pune Section International Conference*, Pune, India, 14-16 Dec 2023, pp. 1-6.
- [3] S Deepthi, K Vijeth, R Rishitha, S Sudarshan and K Vineeth, "Smart Reader for Blind People," *International Conference in Advances in Power, Signal, and Information Technology*, Bhubaneswar, India, 23-25 May 2025, pp. 1-4.
- [4] K Nikhil, Om Sharan and J Manikandan, "Design and Development of a Digital Scribe for Visually Challenged Students," *IEEE Bangalore Humanitarian Technology Conference*, Vijayapur, India, 08-10 October 2020, pp. 1-4.
- [5] Z Shi, Y Tamura, Z Liao, W He and Y Hirata, "Robotic Shopping Guidance System for the Visually Impaired Users Using Servo Brakes*," *IEEE International Conference on Advanced Intelligent Mechatronics*, Boston, MA, USA, 15-19 July 2024, pp. 1416-1421.
- [6] K Nunokawa, T Tanabe, H Takizawa, K Doi and S Ino, "Navigation for precise walking of blind people based on pulling illusion devices embedded in white canes," *4th Global Conference on Life Sciences and Technologies*, Osaka, Japan, 07-09 March 2022, pp. 235-238.
- [7] M Y Wang, T J Fang, L J Kau, W D Yeh and J S Sun, "A Smart Saddle for the Status Monitoring of Guide Dogs," *International Conference on Consumer Electronics - Taiwan*, Taichung, Taiwan, 09-11 July 2024, pp. 771-772.
- [8] Y Chen *et al.*, "Quadruped Guidance Robot for the Visually Impaired: A Comfort-Based Approach," *IEEE International Conference on Robotics and Automation*, London, United Kingdom, 29 May – 02 June 2023, pp. 12078-12084.
- [9] H Abdelmoneim, A Elwakil and M Moussa, "Vision Assist System (VAS): A Smart Cane for the Visually Impaired," *IEEE International Conference on Smart Sustainable Systems for Computer and Engineering Applications*, Cairo, Egypt, 19-21 Apr 2026, pp. 1145-1149.
- [10] S R Robotics: <https://srrobotics.in/product/smart-glasses-for-blind/>
- [11] Jyoti-Ai glass: <https://jyoti-ai.com/>
- [12] Vision Aid India: <https://visionaidindia.org/about-smart-vision-glasses/>
- [13] SMARTON: <https://www.amazon.in/SMARTON-Smart-Glasses-Visually-Impaired/>
- [14] eSight Eyewear: <https://www.esighteyewear.com/>
- [15] A smart eyeglass device for assisting a visually impaired user and its method thereof. Patent Application No. 202641021849A dated 24-Feb-2026. Published in Indian Patent Office Journal dated 13/03/2026 Page No. 30325.